

Using AI Applications to Achieve Environmental Sustainability in North Africa

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In this module, Maha and Muhammad explore the ethical use of AI technologies to address climate change in North African countries. Looking at examples from Mauritania, Libya, Egypt and more, they highlight some of the challenges and contexts specific to the region before pointing to potential solutions. They conclude their module with a series of recommendations around implementing AI systems in the region, such as the use of ethical frameworks to promote inclusivity and protect vulnerable populations.

Note: Please refer to the presentation in the additional sources for more information on the examples and use cases from this lecture.

Lecture Transcript

Jouini 0:00 Again, *bonjour* from Mauritania. Our lecture today is about using AI applications to achieve environmental sustainability in North African countries. The research was launched and led by Maat for Peace, Development and Human Rights, one of the leading organizations in promoting development and defending human rights in North Africa. They are based in Egypt and Maat is the name of the gods of ethics at the Egypt's old time. And from the name of the organization, we can tell that Africans are ethical. We are ethical. That's why today when we talk about AI ethics, and climate change, we have to start from ourselves. And in partnership with African Center for AI and Digital Technology, a pan-African think tank based in Mauritania.

Jouini 1:13 Muhammad Mokhtar, my colleague, he doesn't present himself. Muhammad Mokhtar is the deputy director of the Democracy and Good Governance Unit at Maat for Peace, Development and Human Rights. He is a human rights defender and expert in political technology. Also, Muhammad wrote many research pieces, and he spotlights on the impact of AI on human rights and society. Maha Jouini, I'm an author and founder of Africa Center for AI and Digital Policy. I'm based in Mauritania. I hold a Master's in Applied Technology from Tianjin University of Education and Technology in China. And now I'm continuing my PhD. Also, I'm the regional coordinator at the Global Responsible AI Hackathon 2023, launched by Women in AI Ethics. I'm a speaker and an analyst.

Mokhtar 2:14 Talking in this course about using artificial intelligence applications to achieve environmental sustainability in North African countries. Okay, first of all, we introduce the environmental crisis in North Africa, in Egypt, in Algeria, in Libya, in Tunisia, in Morocco, and in Mauritania, okay. First of all, for example,

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the World Health Organization expected that the climate change will kill 250,000 people per year between 2030 and 2050, okay, due to diseases associated with climate change phenomena, such as malaria, and heart diseases, okay. And also in the midst of 2030s, climate change will push more than 130 million people into poverty. In addition to placing 200 million people in conditions that is wanting to receive the emergency human assistance, okay.

Mokhtar 3:39 This has an effect on human rights in these countries. A lot of people will suffer from environmental problems like high rates of air and water pollution, in addition to climate change, and the catastrophe that is related to the following falling and the lack of fair information and something like this. In this era that is artificial intelligence applications will play an important tool to overcome these problems and to you know, to overcome these problems and to give the people that mechanism to combat the bad effects of the environmental catastrophe and human rights. Okay.

Mokhtar 5:04 This course will discuss six parts. The first one Introduction. The second one is threats related to environmental diseases in North Africa, and the third one is the role of artificial intelligence application in achieving sustainable environmental, environmental sustainability. We will specifically discuss how artificial intelligence overcomes environmental disasters in many countries okay. And after that in in the fourth session, we will discuss the successful experience of using artificial intelligence to achieve sustainable environmental development. After that we will discuss as a chain you know, challenges in implementing an artificial intelligence technology in this era, okay. And this and finally, we will discuss recommendations, okay. In the introduction, we will discuss the widespread threat of environmental crisis in human rights around the world and also we'll discuss that climate change will have an effect on people's health, on people's lives, and effects on economics, human rights and economic and social human rights of the people okay. And also we will discuss that numbers about people will affect in the futures and in the futures is that environmental crisis will affect on that social life is of the people. It will kill 250,000 people between 2020 and 2030 okay?

Mokhtar 7:24 And especially in North Africa, we will discuss the houses and streets that will be affected by the spirit right now. Okay, the environmental related to disasters is high rates of drought and the higher and the floods, the air and water pollution okay, they all have this affected on human rights through North Africa okay. This phenomenon affected to people's live is due to the water shortage, food insecurity, increased rate of disease and also that this death caused by raising that feeling of temperatures. The affected change of climate change on infrastructures and the poor people, the oldest people, the youngest people and the children, ok, are all affected on these are human rights, as shortage of the foods, the shortage of the water (there is no clean water) and also we will discuss the pollution. Okay.

Mokhtar 8:56 We will beginning with Egypt because there is an important environmental issues with water scarcity in Egypt and the air pollution and rising sea levels okay. And the climate change will affect it on the people lives also in Egypt and ecosystem. And after that, we will continue to explain how Morocco will be affected by the climate change and also by the environmental disasters like scarcity of the water and also the air pollution and problems related to problems related to that ecosystem okay, in Morocco. For example, the air pollution causes more than 5,000 deaths in Morocco and costs the state 11 billion dirhams annually okay. And this all affected on the whole sector - the health care and the food and the people. And we will discuss Tunisia and how climate change and environmental disaster will affect it on Tunisia and Libya and Mauritania. After this we

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will discuss the role of Artificial Intelligence application in achieving environmental sustainability okay because of an around the world with many, many stakeholders and the cities, civil societies groups take out steps to combating environmental disasters by using the artificial intelligence applications. And these uses have a lot of other uses of artificial intelligence applications in other ways. Number one, it would be in your warnings system and you will give a lot of examples of this. And after that, it will be improving the food security, improved service to the people affected by environmental threats and also reduce greenhouse gas levels, monitoring air pollution and preventing poaching, that is harm is environmental, and supporting biodiversity okay.

Mokhtar 12:13 And we will discuss after this okay, how we can, how we can use these applications and achieve sustainable environmental development in North Africa and the challenges that will face these countries to use artificial intelligence or artificial applications in combating environmental disaster. Okay and after that we will finish by recommendations. Okay. So that's our course. I finished my presentation right now and I will look forward to know your opinions and your adding in value. And also I wanted to listen to Dr. Maha. Okay - thank you.

Jouini 13:20 Muhammad he covered it - the environmental crisis in North Africa. And he also covered the threats related to the environmental disasters in our region. And we highlighted the situation in Egypt, Morocco, Tunisia, Libya. And I would love to add only one one information about Libya, because the climate change in Libya puts Libya in very complicated situations because people, they are suffering from the land degradation and sustainable energy production, air and water pollution and the loss of biodiversity areas are of a global concern. Moreover, Libyans are suffering from war, are facing the damage of civil war in Libya. In this results, people are migrating from the rural region which is supposed to be green and moving to the capital, moving to Tripoli. So in this case, we can say that the climate change is putting a lot of complication at Libyan context. According to ACRC, and the United Nation mission in Libya, they say that there are in Tripoli, in the west of Tripoli, there are a militia group who took over the places where there are water and green spaces. They took it from their owners and they pushed them to go away from it. So now in Libyan context, climate change is the factor, is another factor, to increase the rate of civil wars between Libyans.

Jouini 15:51 Also in Mauritania, and here in Mauritania, there are something which we have to highlight that the country already is suffering from desertification, from deforestation, and so on. And also, the capital, Nouakchott is ranked on the list of the 10 cities most vulnerable to drowning due to climate change, as most of its neighborhoods are located below the water level of the Atlantic Ocean along with this dilapidated and fragile infrastructure. On the other hand, there is an increase of desertification and drought rates, which leads to high rates of displacement from regions and various rural area. And when I talk about displacement, I have to mention that displaced people - they are generally the children of the free slaves. And most of them, they come from nomadic group and some vulnerable groups. As results of their looking for water and green space, we have to understand that the most affected groups, they are the marginalized group at the origins. At the end of the day in case of Mauritania, we can understand that climate change is enhancing and making the social inequality more and more complicated. The marginalized groups, some of the black community, the children of the free slaves, they will become by the time more and more vulnerable, and they were, they were still living in the displacement to look for, for green spaces and clean water to drink.

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Jouini 18:20 Here we go, the role of artificial intelligence in achieving environmental sustainability. Many stakeholder with their states, civil society groups or corporations have realized the disappointed situation that humanity has reached due to the successive environmental crisis, and have begun to use many innovative ways to combat the sequences of the most harmful environmental crises leading to a reduction in human rights violations resulting from them. Among the most prominent of these methods was the use of AI techniques to protect the environment and conserve resources with the as early warning system by predicting and preparing for the disaster related to extreme climate changes, and predicting the resulting forced displacement or by providing humanitarian aid to those affected by it with assisting the various harmful effects resulting from this crisis or by working to minimize or reduce the levels of greenhouse gas emissions to the closest possible level. The Paris Agreement on climate change indicates the need to reduce emissions by 45% by 2023 and reach net zero emissions by 2050, or by monitoring air and water pollution at just-efficient operation at the four stations.

Jouini 20:21 On all the preservation of the health of the oceans as well as biodiversity at the same time. At the same time, they call to implement and using artificial intelligence to design basic infrastructure capable of dealing with environmental crisis. Like Mokhtar said, he has already mentioned the early warning system and these pictures come from India. This is an early warning system about inundations (floods). In the midst of this, our AI system plays a key role as early warning systems to predict natural disasters related to climate change. For example, floods cause severe damage to the infrastructures, in addition to the death of many citizens and the displacement of families. Also, monitoring flood operations on a continuous basis helps governments to determine their severe impacts, and plan for relief operations. In this way, the Flood Forecasting Initiative, in which Google cooperates with many governments is working to protect citizens from flood damage as much as possible by developing system based on AI that predicts when and where floods occur. If this has happened, it will improve prove the lives of more than two millions people who suffer from floods in India, and 40 million people who suffer from the same matter in Bangladesh.

Jouini 22:27 Also, we have to improve food security. In this context, many applications of AI are working to improve food security related to climate change, with the wide availability of high resolution image from satellite, and aerial based on AI, it has become possible to monitor crop conditions and agricultural productivity on a large scale and provide services to those affected by environmental threats. All technology provides services to those affected by climate disasters and assess damages arising from climate disasters. AI technology are integrated into the analysis of floods use satellite imagery to report rapid response and assesses and compensate for various harm.

Jouini 23:31 Reduce greenhouse emissions level. AI applications reduce greenhouse gas emissions level and reduce energy damage in various industry. Many companies use AI to help steel plants reduce their industry CO2 emission by four, five hundred sorry for 4500 tons per year. Many estimators and studies suggest that AI based machine learning is a powerful tool to reducing greenhouse gas emissions. Also monitoring air pollution. Many AI application are very good at monitoring air pollution, improving its quality, and monitoring poaching, deforestation, smart air purifiers using AI can record air quality and environmental data moment by moment by moment and work to increase the efficiency of air filters, prevents poaching that harm the environment. AI is used to prevent poaching that harms the environment in Cambodia as example AI program based on machine learning ways design it to predict poaching to reduce it and protects wildlife, support biodiversity, and preserves the health of the oceans.

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Jouini 24:55 AI system supports biodiversity and preserve the health of the ocean. AI is used to detect change in biodiversity around the world and collect data from difficult or inaccessible ocean sites to maintain ocean health. Successful experience using AI to achieve sustainable environmental developments. Many countries, civil society organizations and companies have successfully used AI application to promote environmental conservation and combat climate change through different successful experience are viewed below to identify and build on the reasons for their successful in building other experience in the future. It can present this experience in detail as follows. For example, in Georgia, the government has established an integrated climate change a story has the government has established an integrated climate risk management system in cooperation with UNDP and substantial funding from the Green Climate Fund. AI was used to monitor extreme climate conditions in their system, especially those related to temperature increases using machine learning models to predict disaster related to climate change, and many respects are developing to deal with them. The algorithm developed by UNOSAT is used to detect climate change related floods in many regions across the world and detect rains associated with the climate change. Also the model was used in many Myanmar, Mozambique, and many regions where climate disaster are continuously frequent. Favorable climates conditions allowed best probably proliferation in East Africa, Southwest Asia and the area around the Red Sea, such as the Locust proliferations associated with climate change in Kenya and many Africans country.

Jouini 27:07 Leverage international expertise that has used AI to achieve environmental sustainability and its application to North Africa. With regard to the problems of air pollution experienced by many countries, mainly Algeria and Egypt, the use of smart air purification that works to use AI to record air quality and environmental data should be utilized at the moment's notice and work to increase the efficiency of different air filters. Regarding natural disaster associated with climate change that threatens Libya, Algeria, and the coastal area of Egypt and Mauritania AI system should be used in an early warning to predict and try to reduce such disaster as well as to enhance the humanitarian response of those affected satellite technology may be must be used to predict floods that may hit Morocco, floods that may hit Egypt constantly. In terms of reducing greenhouse gas emission levels as well as energy damage, damage in different industry, it is possible to use in house sensor and smart meters to collect data, monitor and analyze and improve energy in us building tool to reduce greenhouse gas emissions to preserve the environments. Regarding detection of changes in biodiversity, the use of AI robots to monitor ocean and lake conditions such pollution level, temperature, temperature and pH should be studied, especially in the coastal cities of Morocco, Tunisia and Algeria.

Jouini 28:54 Challenges in implementing AI technology. The top of these challenges is the need of funding and various funds to invest in AI applications related to environment sustainability. Also, in addition to the increasing of internal conflicts and terrorism, especially in the case of Libya. A lack of technology awareness, especially the applications of AI among all governments agencies within the countries of North Africa, in conjunction, continue with US policy of measure stance that exasperate climates problem without considering their impact on citizens in other regions of the world. Week UN and human rights bodies in dealing with issues of integrating new technology into the fight against pressing environmental threats. With health system infrastructures, an early warning system depends on AI, even the general culture of citizen to respond effectively to climate change. Lack of political will and political willingness to change and using AI applications to achieve environmental sustainability and North African countries. Recommendations, I would love to give the floor to Mokthar for the recommendations.

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Mokhtar 30:22 Okay, I am with you right now, okay. There are a lot of recommendations. The first one is we need to rely on all artificial intelligence to predict about the environmental disasters and also we need to use artificial intelligence application in fighting against air and water pollution especially in Egypt, okay and in Morocco also. Also we need to rely on artificial intelligence application to reduce greenhouse gas emissions okay, levels that caused extreme weather okay because this phenomena will reduce the food okay and will reduce the food and the water. And it will be a short food shortage and water shortage. And also we need to strengthen collaboration between North African countries with regard to research and study related to the promotion of artificial intelligence in fighting against environmental disasters, okay. And sharing experiments among them and strengthen cooperation in this regard, because there are shortage of cooperation between the states. And we need the civil society and states to really cooperate in this field to achieve a good cooperation and in combating the environmental disasters really relying on artificial intelligence. Okay. Also in North Africa, policy efforts have to be built as secure infrastructure to support the local development of artificial intelligence, and empowering the policymaker in this field to understanding as the use of technology and the use of artificial intelligence in combating this phenomenon. Finally, we should advocate for the responsible use and development of artificial intelligence in Africa and urging African governments to launch African artificial intelligence ethics guideline to use it. And thank you all of you all.

Jouini 33:34 Thank you Mokhtar. Thank you. This is the end of our lectures. And we ended to stress to the importance of artificial intelligence, AI, ethics guidelines, because we believe that AI should be used to reduce climate change certainly, and also, it has to be used for the common good and better future for vulnerable groups, marginalized groups, poor women, and all of minorities in the world wide. AI should be used for inclusivity and the common good.